

12
21
35
68

CLUSTER UNIVERSITY SRINAGAR

Examination: - B.Tech. 6th Sem
Paper Title: Analysis and Design of Algorithms
Max. Marks: - 100

Session: 2024
Code: CSE-601
Min. Marks: - 40

Branch: CSE
Batch: 2018-20
Time: - 03 Hours

Section - A [Long Answer Type Questions]

35 / 12 x 04 = 48 Marks

- Q1. Use the Division method and Quadratic Probing for resolving collisions to store the values 3, 2, 9, 6, 11, 13, 7, 12 in a hash table of size 10 and hash function $h(k) = 2k + 1$, calculate the average probes, and create a new order?

(OR)

Find the complexity of the following recurrence relation:

- (a) $T(n) = 2T(n-1) - 1$, if $n > 0$,
1, otherwise (use Substitution method)

- (b) $T(n) = 4T(n/2) + n^2$ (use Master Theorem)

- Q2. Discuss various comparison trees for searching and sorting?

(OR)

- (a) What is big O notation? Find $O(f(x))$ for $3x^3 + 2x^2 + 4x$?

- (b) If $f(n) = n^{1+\sin n}$ and $g(n) = n^{1/2}$, Are the functions asymptotically comparable, Justify?

- Q3. Explain the Merge Sort algorithm, including its divide, conquer, and merge steps. Provide pseudocode or algorithmic steps to illustrate the implementation of Merge Sort. Analyze the time complexity of Merge Sort?

(OR)

State and explain the Greedy Knapsack Problem? Find an optimal solution to the Knapsack instance $n=4$ objects and the capacity of knapsack $m=15$, Profits (10, 5, 7, 11) and weight are (3, 4, 3, 5)?

- Q4. What is Dynamic programming? How is it different from the Greedy technique? Also, give a Greedy solution for the change-making problem, considering the denominations : { 500, 200, 100, 50, 10, 5, 2, 1}?

(OR)

Explain the method of reduction to solve TSP (Travelling Salesman Problem) using branch and bound?

Section - B [Problem Solving / Medium Answer Type Questions]

21 / 08 x 04 = 32 Marks

- Q5. What is an algorithm? Explain characteristics of an algorithm with the help of an example?

(OR)

Discuss the concept of Double Hashing and Rehashing?

Q6. Differentiate between Bigoh and Omega notation with an example? 6

(OR)

Explain the Recursive Binary search algorithm with a suitable example?

Q7. Explain the following, together with their respective significance.

(a) Decidable and Undecidable problem.

(b) Deterministic and Non-Deterministic Algorithm.

(OR)

Show a polynomial time reduction from the Clique problem to the Vertex Cover problem by giving an example?

Q8. Explain any one application of Backtracking?

(OR)

Compare NP-hard and NP-completeness?

Section - C [Very Short Answer Type Questions Attempt All]

12/04 x 05 = 20 Marks

Q9. How to choose Hash Function?

Q10. Find the lower bound of the running time of the quadratic function

$$f(n) = 3n^2 + 2n + 4.$$

Q11. Show $\sqrt{6n^3 + 7n^2 + 3n + 5} \in O(n^{1.5})$

Q12. What is a minimum spanning tree?

Q13. What is the principle of optimality?

$$3n^2 + 2n + 4$$

$$f(n) \geq c(g(n)).$$

$$3n^2 + 2n + 4 \geq 3n^2$$

$$3 \cdot n^2$$

$$c = 3.$$

$$P(n) \rightarrow 1/n^2$$

CLUSTER UNIVERSITY SRINAGAR

Examination: - B.Tech. 6th Sem
Paper Title: Advanced Microprocessors
Max. Marks: - 100

Session: 2024
Code: FEE-610
Min. Marks: - 40

Branch: CSE
Batch: 2018-20
Time: - 03 Hours

Section - A [Long Answer Type Questions]

12 x 04 = 48 Marks

- Q1. Draw the internal architecture of 8086 microprocessor. Discuss its Execution Unit (EU) in detail.

OR

Define addressing mode. Discuss different addressing modes supported in 8086 microprocessor.

- Q2. Discuss interrupt structure in 8086 microprocessor in detail.

OR

Discuss in detail different memory and input-port port addressing techniques in microcomputer systems.

- Q3. Draw the internal architecture of 8087 numerical data processor. Discuss its Control Unit (CU) and Numeric Extension Unit (NEU).

OR

Discuss the keyboard/alphanumeric display interfacing with microprocessor in detail.

- Q4. Discuss the different functional blocks in 80186 microprocessor architecture.

OR

List the functions of Bus Interface Unit (BIU). Discuss the pipelining in 8086 in detail.

Section - B [Problem Solving / Medium Answer Type Questions]

08 x 04 = 32 Marks

- Q5. Discuss the flag register in 8086 microprocessor.

OR

Briefly explain different loop and jump instruction in 8086 microprocessor.

- Q6. Give brief description of different 8086 microprocessors signals.

OR

Discuss 8086 microprocessor read and write machine cycles with necessary timing diagrams.

- Q7. Explain interfacing of 8087 co-processor with 8086 microprocessor.

OR

Explain briefly different operating modes of programmable peripheral interface 8255 IC.

- Q8. Give brief account of 80186 microprocessor family.

OR

Write short note on Pentium IV processor.

Section - C [Very Short Answer Type Questions Attempt All]

04 x 05 = 20 Marks

- Q9. List the differences between 8086 and 8088 microprocessors.

- Q10. Explain briefly memory segmentation in 8086 microprocessor.

- Q11. List the main features and data types supported by 8087 numerical data processor.

- Q12. Explain any two arithmetic instructions of 8086 microprocessor.

- Q13. Difference between Loosely Coupled and Tightly Coupled Multiprocessor System.

CLUSTER UNIVERSITY SRINAGAR

Examination: - B.Tech. 6th Sem
Paper Title: RDBMS
Max. Marks: - 100

Session: 2024
Code: CSE-602
Min. Marks: - 40

Branch: CSE
Batch: 2018-20
Time: - 03 Hours

Section - A [Long Answer Type Questions]

12 x 04 = 48 Marks

Q1. Describe the components of a DBMS and their functions. Discuss the role of each component.

(OR)

Explain the Entity-Relationship (ER) model in detail. Describe its components such as entities, attributes, and relationships, and illustrate how they are represented diagrammatically.

Q2. Discuss the significance of logical and physical pointers in file organization. Explain how pointers are used to navigate through records and access data efficiently in different file organization techniques

(OR)

Describe the Direct Files organization technique. Discuss its structure, advantages, and limitations compared to other file organization methods.

Q3. Explain the concept of keys in the relational model. Discuss primary keys, candidate keys, and foreign keys, and their significance in maintaining data integrity

(OR)

Explain the process of normalization using functional dependency. Discuss the various normal forms such as 1NF, 2NF, 3NF, BCNF, and their importance in database design

Q4. Discuss the techniques for reflecting updates to the database and recovery in distributed database systems. Explain how distributed transactions are managed and recovered in the event of failures

(OR)

Discuss the concept of virtual memory in database systems. Explain how virtual memory management techniques are used to optimize memory utilization and improve system performance

Section - B [Problem Solving / Medium Answer Type Questions]

08 x 04 = 32 Marks

Q5. Describe the concept of data independence and its types.

(OR)

Compare and contrast the Relational and Hierarchical data models.

Q6. Explain briefly the Direct file organization technique along with its limitations as compared to other file organization methods.

(OR)

Discuss the concept of record placement and its impact on file access performance.

Q7. Discuss the basic principles of Relational Algebra. Give examples.

(OR)

Explain the basic SQL commands for retrieving, inserting, updating, and deleting data from relational databases.

Q8. Discuss the properties of transactions in database management.

(OR)

Discuss the advantages and disadvantages of using a DBMS in distributed databases.

Section - C [Very Short Answer Type Questions Attempt All]

Q9. What is the role of data modelling in database design?

Q10. What is a view in DBMS?

Q11. Differentiate between Serial Files and Sequential Files.

Q12. Define join dependencies in database normalization.

Q13. What are the key principles of disaster recovery?

3
4
2
3
2

28

17

10

55

CLUSTER UNIVERSITY SRINAGAR

Examination: - B.Tech. 6th Sem
Paper Title: Operating Systems
Max. Marks: - 100

Session: 2024
Code: CSE-604
Min. Marks: - 40

Branch: CSE
Batch: 2018-20
Time: - 03 Hours

Section - A [Long Answer Type Questions]

12 x 04 = 48 Marks

- Q1. Explain the functions and characteristics of an operating system in detail. How does an operating system manage hardware resources effectively?

(OR)

Explain the concept of spooling in operating systems. How does spooling work, and what advantages does it offer in managing I/O operations?

- Q2. Explain the concept of process scheduling. Discuss the different scheduling criteria used in process scheduling and how they impact system performance.

(OR)

Explain the concept of semaphores and their role in process coordination. Discuss the implementation of semaphores and how they can be used to solve synchronization problems.

- Q3. Discuss deadlock avoidance strategies in operating systems. Explain how these strategies dynamically allocate resources to avoid deadlocks and discuss their advantages and limitations.

(OR)

Explain the principle of locality and its relevance to cache memory. Describe briefly the cache replacement policies such as Least Recently Used and Random Replacement.

- Q4. Explain essential UNIX commands used for file management, navigation, and process handling. Provide examples of commonly used UNIX commands and their functionalities.

(OR)

Describe the process creation mechanism in UNIX. Explain how new processes are created in UNIX systems and discuss the role of the `fork()` and `exec()` system calls.

Section - B [Problem Solving / Medium Answer Type Questions]

08 x 04 = 32 Marks

- Q5. How virtual machines are implemented and discuss their significance in operating system environments.

(OR)

- Q6. Define race conditions and explain why they occur.

(OR)

Describe the critical section problem and its impact on concurrent processes.

- Q7. Define memory management in operating systems. Explain the difference between logical and physical address space and discuss their respective roles.

(OR)

Define segmentation in memory management. Discuss the benefits of segmentation and explain how it differs from paging.

- Q8. Compare and contrast the Bourne shell and C shell in UNIX. Discuss their features, syntax, and usage in shell scripting.

(OR)

Explain the concept of trapping signals in UNIX. Discuss how signals are handled in UNIX systems. Provide examples of common signals.

P.T.O

Section - C [Very Short Answer Type Questions Attempt All]

04 x 05 = 20 Marks

~~Q9.~~ Define the operating system. Mention two characteristics of an operating system. 3

~~Q10.~~ Define the term "multiprogramming." 3

~~Q11.~~ What is demand paging? 3

~~Q12.~~ Enlist the necessary conditions for a deadlock to occur. illustrate with an example. 2

~~Q13.~~ What are the responsibilities of a system administrator in UNIX?

1
11
17
27
55

CLUSTER UNIVERSITY SRINAGAR

Examination: - B.Tech. 6th Sem
Paper Title: JAVA Programming
Max. Marks: - 100

Session: 2024
Code: CSE-603
Min. Marks: - 40

Branch: CSE
Batch: 2018-20
Time: - 03 Hours

Section - A [Long Answer Type Questions]

12 x 04 = 48 Marks

- Q1.** Explain Java Looping Statements in detail with the help of syntax and program?
(Hint: Syntax and program of all the four types of loops: while, do-while, for, for-each loop must be discussed)

(OR)

What is a token? Explain in detail the various classifications of tokens. Write a program to determine whether a person is eligible to vote or not using the ternary operator. (2, 7, 3)

- Q2.** Define a Class in Java? WAP to justify the use of Class in Java? Discuss Static members? (3, 5, 4)

(OR)

What is Inheritance? Discuss different types of Inheritance supported by Java with the help of program of each Inheritance type? (3, 9)

- Q3.** Discuss Exception handling in detail? What are different ways to handle Exception in Java? (5, 7)

(OR)

What is Applet? Discuss Applet life cycle? Write all the steps for the execution of Applet program?
(Hint: Create an HTML extension file for the execution of the Applet program) (2, 4, 6)

- Q4.** Discuss Abstract Windowing Toolkit (AWT)? Explain various layout managers of AWT? (3, 9)

(OR)

Discuss user interface components in AWT? WAP to print the basic form using AWT user interface? (8, 4)

Section - B [Problem-Solving / Medium Answer Type Questions]

08 x 04 = 32 Marks

- Q5.** Discuss Decision making statements of Java in detail? Explain syntax and program with the help of output of different if-statements? (3, 5)

(OR)

Explain in detail the various features of Java?

- Q6.** Discuss One-Dimensional Array and Two-Dimensional Array with the help of program?

(OR)

Discuss Method overloading and Method overriding with the help of the program?

- Q7.** What is a thread? Discuss life cycle of a Thread?

(OR)

Write a program to implement thread using a runnable Interface?

- Q8.** How are frames and panels useful in Java AWT?

(OR)

What are the key components of Java AWT?

Section – C [Very Short Answer Type Questions Attempt All]

04 x 05 = 20 Marks

- Q9. Briefly discuss Command line Arguments in java with the help of program? ↵
- Q10. How Java language is different from C and C++ language? ✓
- Q11. How wrapper class is useful for primitive datatypes?
- Q12. What is the purpose of the finally clause of a try-catch-finally statement?
- Q13. Briefly discuss Java AWT Canvas? ✓

CLUSTER UNIVERSITY SRINAGAR

Examination: - B.Tech. 6th Sem
Paper Title: Software Engineering
Max. Marks: - 100

Session: 2024
Code: CSE-605
Min. Marks: - 40

Branch: CSE
Batch: 2018-20
Time: - 03 Hours

Section - A [Long Answer Type Questions]

12 x 04 = 48 Marks

- Q1. Draw and Describe Spiral Model of Software Development in detail, also list its merits and demerits.
(OR)

Discuss the concept of Layered Technology in Software Engineering. How does the Layered approach promote abstraction, modularity and reusability in Software Development?

- Q2. Briefly describe COCOMO Model and use it to calculate the following:

i). Effort ii). Development Time iii). Average Staff Size iv). Productivity

If the estimated size of the project is 200000 LOC using embedded mode.

(OR)

What are the different phases of the project management process, and how does each phase contribute to the successful execution of a project? Provide illustrative examples for each phase to demonstrate its practical application.

- Q3. What is meant by Software Quality? Explain in detail various Software Quality factors.

(OR)

What is Software Quality Assurance (SQA)? Discuss in detail various SQA activities.

- Q4. What is Black Box Testing? Explain the different types of Black Box Testing strategies by giving suitable examples.

(OR)

What is White Box Testing? Explain the different types of White Box Testing strategies by giving suitable examples.

Section - B [Problem Solving / Medium Answer Type Questions]

08 x 04 = 32 Marks

- Q5. Explain the concept of Process Framework in Software Engineering and also describe Process Framework Activities.

(OR)

What are the Legacy Software Systems? How do they contribute to challenges in Software Development?

- Q6. Discuss the process of Software Project Scheduling and also list its key Components and Techniques.

(OR)

Discuss in detail key principles of Effective Modular Design in Software Engineering.

- Q7. Discuss McCall's quality factors in Software Engineering.

(OR)

Discuss the concepts of Transform and Transaction mapping in Software Engineering.

- Q8. Elaborate on the concept of Software Maintenance and its types.

(OR)

Explain the key differences between Reverse Engineering and Re-Engineering?

Section – C [Very Short Answer Type Questions Attempt All]

04 x 05 = 20 Marks

~~Q9.~~ What are the key attributes of good Software? ✓

Q10. Briefly describe some common Software Myths. ✓

~~Q11.~~ What are the key differences between System and Integration testing methodologies? ✓

~~Q12.~~ What is Software Reliability and why is it important? ✓

Q13. Briefly Explain Software Risk Analysis. ✓